CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (CBIT)

Department Of Computer Science And Engineering

FUNDAMENTALS OF DATA SCIENCE (22CSC59)



CASE STUDY REPORT

Title: East-West Airline
and
Wine Quality Assessment

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Case-study-2: East-West Airline

1. Problem statement and objective

East-West Airlines seeks to understand customer differences in frequent-flyer activity, reward earning and use, credit-card miles, balances, and recent flight behavior. The objective is to identify useful customer segments using unsupervised clustering and interpret what distinguishes each segment. The supplied workbook contains the customer records and a description sheet noting that the company name was changed.

Dataset audit

Item Finding

Customer records 3,999

Columns in data sheet 12

Missing cells 0

Exact duplicate rows 0

Duplicate customer IDs 0

Clustering inputs 10 behavioral variables; ID# excluded; Award? reserved for profiling

Methodology

- Excluded ID# because identifiers do not describe behavior.

defining the segments directly by redemption status; it is used later to profile the segments.

- Applied log1p to highly skewed balance, qualification-mile, bonus-mile, transaction, recent-flight, and

enrollment-duration variables.

- Capped transformed skewed variables at their 1st and 99th percentiles, median-imputed missing values (none were

present), then used RobustScaler for distance-based methods.

- Compared K-Means for k = 2-8, hierarchical linkage variants, and DBSCAN across multiple eps/min_samples

settings. Evaluated with silhouette and other internal metrics where applicable.

Preprocessing rationale. Extreme frequent-flyer customers may be legitimate high-value customers. Capping reduces

their distance leverage without deleting their records; the trade-off is that the most extreme differences are

compressed. Results can change with alternative scaling or feature choices.

2. Statistical findings, EDA, and K-Means selection

Figure 1. Balance and bonus-mile distributions after log1p transformation for visualization. Both remain concentrated toward lower

values with a long upper tail, supporting the use of a skew treatment and robust scaling before distance calculations.

K-Means parameter evaluation

k Inertia Silhouette Davies-Bouldin Cluster sizes

2 18962 0.659 0.561 3773, 226

3 13164 0.360 0.939 2857, 916, 226

4 9761 0.354 0.998 719, 1973, 226, 1081

5 7980 0.330 0.969 724, 1246, 466, 1337, 226

6 7251 0.328 1.095 1179, 1065, 226, 467, 563, 499

7 6704 0.304 1.227 883, 464, 226, 439, 939, 577, 471

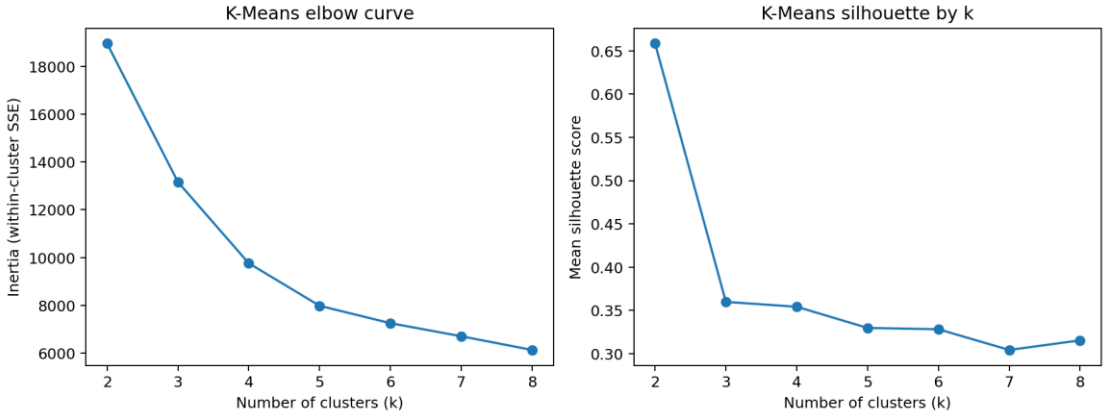
8 6130 0.315 1.113 815, 226, 429, 463, 874, 512, 230, 450

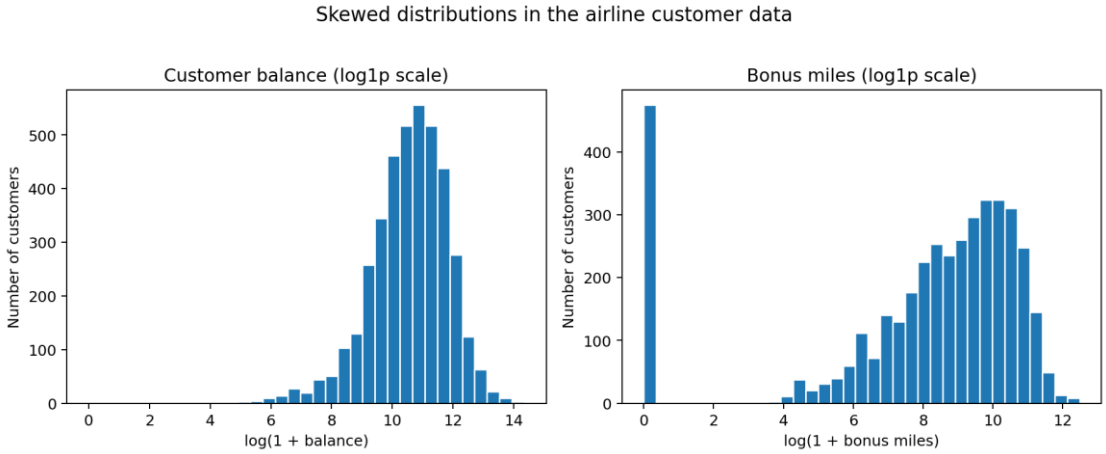
Figure 2. Inertia decreases as k rises, as expected; the silhouette score is highest at k = 2 (0.659). A two-cluster solution isolates the

226 customers with stronger recent flight activity from the much larger remainder. k = 3 is used as the operational profile because it

separates an additional low-reward-activity segment, but its lower silhouette (0.360) means its structure is less clean and should be

validated with the business team.





3. Model evaluation: Hierarchical clustering and DBSCAN

Hierarchical clustering was evaluated with Ward, complete, and average linkage at k = 3 and k = 4. DBSCAN was

tested using eps values from 0.75 to 2.50 and min_samples values of 5 and 10. For DBSCAN, silhouette is computed

on non-noise records only; noise points are not treated as one coherent cluster.

Selected algorithm comparison

Algorithm Clusters Silhouette Davies-Bouldin Noise

K-Means (k=3) 3 0.360 0.939 0.00%

Hierarchical Ward (k=3) 3 0.433 0.692 0.00%

Hierarchical complete (k=3) 3 0.639 0.709 0.00%

DBSCAN (eps=1.5, min_samples=10) 3 0.639 - 0.50%

Ward linkage provides a balanced three-cluster structure (silhouette 0.433). Complete linkage scores higher (0.639),

but isolates a very small 15-customer group, making the segmentation less balanced. DBSCAN with eps = 1.5 and

min_samples = 10 finds three dense clusters and marks about 0.50% of records as noise. These scores should be

compared with cluster sizes and business interpretation, not used in isolation.

DBSCAN parameter sensitivity

eps min_samples Clusters Noise % Silhouette (non-noise)

0.75 5 9 7.55 0.136

0.75 10 4 9.55 0.160

1.00 5 8 2.98 0.020

1.00 10 7 5.35 0.022

1.25 5 4 0.98 0.627

1.25 10 3 1.60 0.634

1.50 5 3 0.40 0.639

1.50 10 3 0.50 0.639

2.00 5 3 0.08 0.639

2.00 10 3 0.40 0.639

2.50 5 2 0.00 0.659

2.50 10 2 0.00 0.659

At eps = 0.75-1.00, DBSCAN generates more small clusters and/or more noise. Around eps = 1.25-2.00, it finds a small set of dense

groups and low noise. At eps = 2.50, the labels collapse to a two-group split. This illustrates why eps and min_samples must be

assessed jointly.

Cluster profile cross-check: Ward and DBSCAN

The alternative algorithms show a similar broad pattern: a core group, a smaller recent-flight-active group, and a

low-reward-activity group. DBSCAN also marks 20 records as noise; those records have unusually high median bonus

miles in this run, so they merit review rather than automatic deletion. Cluster IDs are algorithm-specific and must not

be compared as if they were the same labels.

Algorithm Profile Cluster n Median bonus miles Mean flight miles Award rate

Ward Core reward earners 0 3,311 9,680 438 39.2%

Ward Active flyers 1 226 11,875 1,720 64.6%

Ward Low reward activity 2 462 0 0 8.0%

DBSCAN Noise / unusual profiles Noise (-1) 20 62,914 788 50.0%

DBSCAN Core reward earners 0 3,761 6,654 384 35.3%

DBSCAN Active flyers 1 206 12,898 1,836 69.4%

DBSCAN Low reward activity 2 12 0 0 0.0%

4. Final K-Means segments and interpretation

The k = 3 K-Means solution is used for the descriptive profile below. Segment names are assigned after checking the

raw-data profiles; numerical cluster labels themselves are arbitrary.

Segment profile table

Segment Customers Median balance Median bonus miles Mean flight miles (12 mo.) Award rate

Core reward earners 2,857 56,756 13,495 501 43.3%

Low reward activity 916 14,079 0 20 10.7%

Active flyers 226 66,838 11,875 1,720 64.6%

Figure 3. Active flyers have the highest mean recent-flight mileage (about 1,720 miles) and the highest award rate (64.6%). Low

reward activity has no median bonus miles or bonus transactions and the lowest award rate (10.7%). Core reward earners form the

largest group (2,857 customers), with median bonus miles of 13,495 and an award rate of 43.3%.

Interpretation and possible actions

- Low reward activity (916; 22.9%): lower median balance, zero median bonus miles and bonus transactions, and

low redemption activity. Consider a low-cost re-engagement test, but first confirm whether these customers are still

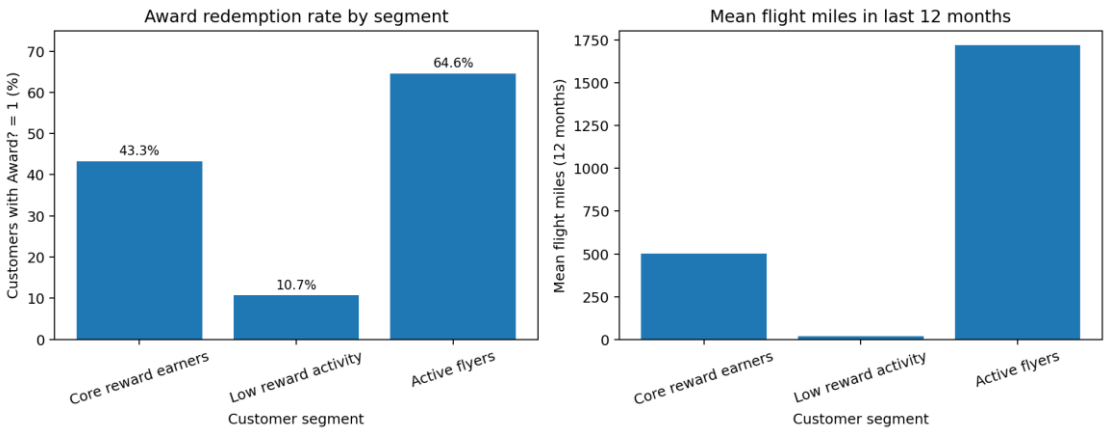
active or eligible.

- Core reward earners (2,857; 71.4%): most customers sit here, with a material reward balance and moderate recent

flight activity. Test personalized reward reminders or credit-card-linked earning offers.

- Active flyers (226; 5.7%): high mean recent flight mileage, a longer median enrollment duration, and high award

rate. Review retention and relevant tier/reward experiences for this group.



5. Visual separation, limitations, and conclusion

Figure 4. Projection of the scaled features onto two principal components, colored by the selected K-Means labels. The view is a 2D

summary of the full feature space, not a complete representation of all dimensions. Overlap in this projection does not by itself

invalidate the clustering.

Model evaluation conclusion

No one algorithm dominates on every criterion. K-Means k = 2 has the best silhouette in the tested k range (0.659),

while k = 3 sacrifices some separation for a more actionable three-segment interpretation. Ward hierarchical clustering

also gives a relatively balanced three-group solution. DBSCAN is useful for surfacing unusual records, but the results

depend on eps and min_samples.

Limitations

- There is no target label against which to validate customer segments; internal metrics measure geometric separation,

not profit or retention impact.

- Capping, log transformation, RobustScaler, and treating the credit-card mile levels as numeric all influence the

distance calculations.

- The snapshot lacks demographics, customer margin, purchase history beyond the supplied fields, and future churn

or retention labels.

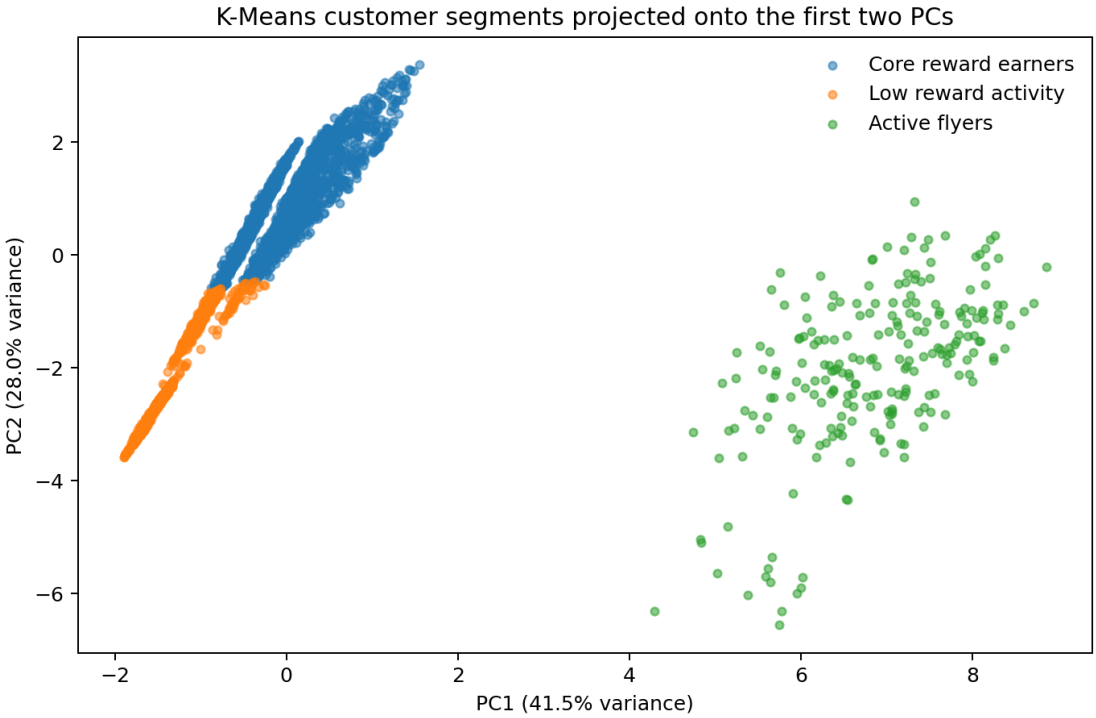
- Award? was excluded from clustering and used only as a profiling variable; segment-level award rates remain

descriptive, not causal.

Conclusion

The supplied dataset supports a defensible exploratory segmentation into low reward activity, core reward earners, and active flyers. These labels can guide hypothesis-driven experiments, but the airline should test response, incremental revenue, retention, and fairness before operational deployment. Submission artifact reminder: the executable notebook and Python script are included in the accompanying package..

Execution url: https://colab.research.google.com/drive/1IgQoZ-8GhcdetvHEL5XyUTmR4KdNthjV?usp=sharing



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Case-study-5: Wine Quality Assessment

1. Problem statement and objective

The company currently relies on sensory quality assessments. This case study investigates whether laboratory measurements can help explain differences in quality scores and support a data-driven screening process. The tasks are descriptive statistics, exploratory visualizations, correlation analysis, multiple linear regression, and business.

Dataset and audit

Item Finding

Red wine records 1,599

White wine records 4,898

Combined records 6,497

Predictor features 11 physicochemical measurements

Missing cells 0

Repeated rows ignoring wine type 1,179

Target Sensory quality score

The red and white files are combined for descriptive analysis. The regression uses exactly the 11 physicochemical

predictors specified in the case brief; wine_type is retained only to audit the source composition and is not a model

predictor. Repeated records are retained rather than arbitrarily deleted. The pooled dataset contains many more white

wine records than red wine records, which affects the overall summaries.

Methodology

- Calculated mean, median, mode, standard deviation, skewness, and coefficient of variation (CV = SD / mean) for alcohol, volatile acidity, and residual sugar.
- Examined Pearson correlation, quality-score distribution, alcohol box plots grouped by quality, and a correlation heatmap.
- Used an 80/20 train-test split with random_state = 42. StandardScaler was fitted on training features only; ordinary multiple linear regression was evaluated on the held-out test set.
- Reported MAE, MSE, RMSE, R-squared, standardized predictor coefficients, pairwise correlations, and variance inflation factors (VIF).

2. **Task 1 - Statistical findings**

Descriptive statistics for the three required attributes

Attribute Mean Median Mode Std. dev. Skewness CV

alcohol 10.4918 10.3000 9.5 1.1927 0.566 11.4%

volatile acidity 0.3397 0.2900 0.28 0.1646 1.495 48.5%

residual sugar 5.4432 3.0000 2.0 4.7578 1.435 87.4%

Relative variability. Residual sugar has the highest coefficient of variation (87.4%), followed by volatile acidity

(48.5%) and alcohol (11.4%). Residual sugar therefore varies the most relative to its mean among these three

variables. Volatile acidity and residual sugar have positive skewness, meaning a longer right tail; their means exceed

their medians.

Alcohol and quality. Pearson r = 0.4443, indicating a moderate positive linear association in the pooled red + white

sample. Higher alcohol content tends to coincide with higher quality scores, but the association is far from deterministic

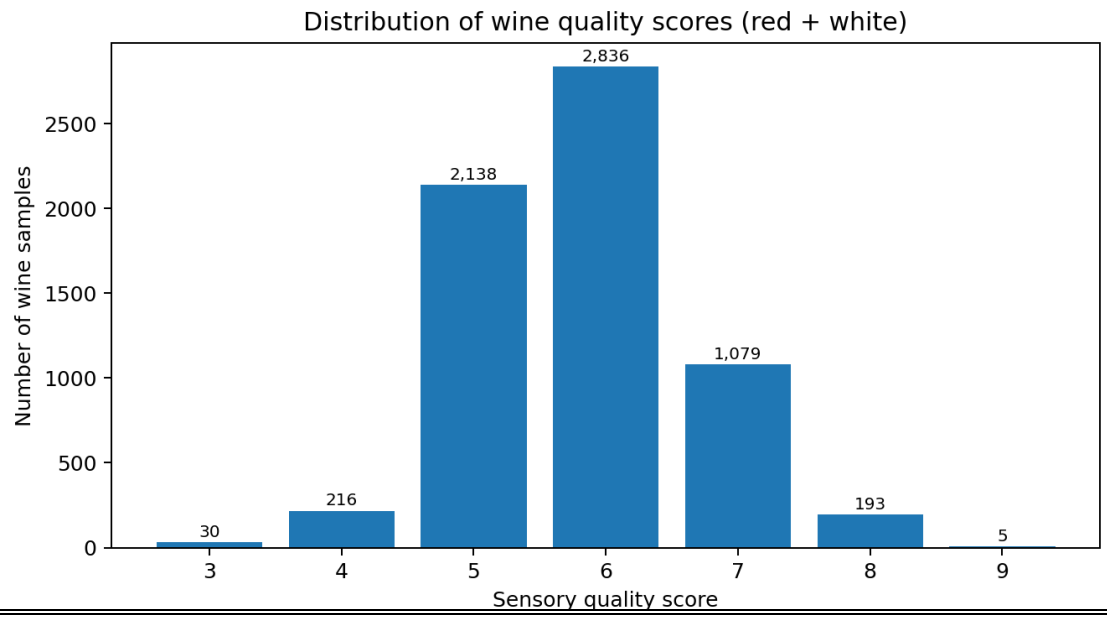
and does not establish causality.

3. Task 2 - Exploratory data analysis

Figure 1. Quality scores are concentrated at 5 and 6: 2,138 samples score 5 and 2,836 score 6. Scores 3, 8 and 9 are rare, and no

score outside 3-9 appears in the supplied combined data. The imbalanced distribution makes extreme quality scores harder to

model.



4. Exploratory patterns and correlation

Figure 2. The median alcohol content generally rises across higher quality scores, consistent with the positive Pearson correlation.

The boxes overlap, however, so alcohol alone cannot reliably distinguish the score of an individual wine. Points beyond the whiskers

indicate potential outlying measurements within a quality group; they are not automatically data errors.

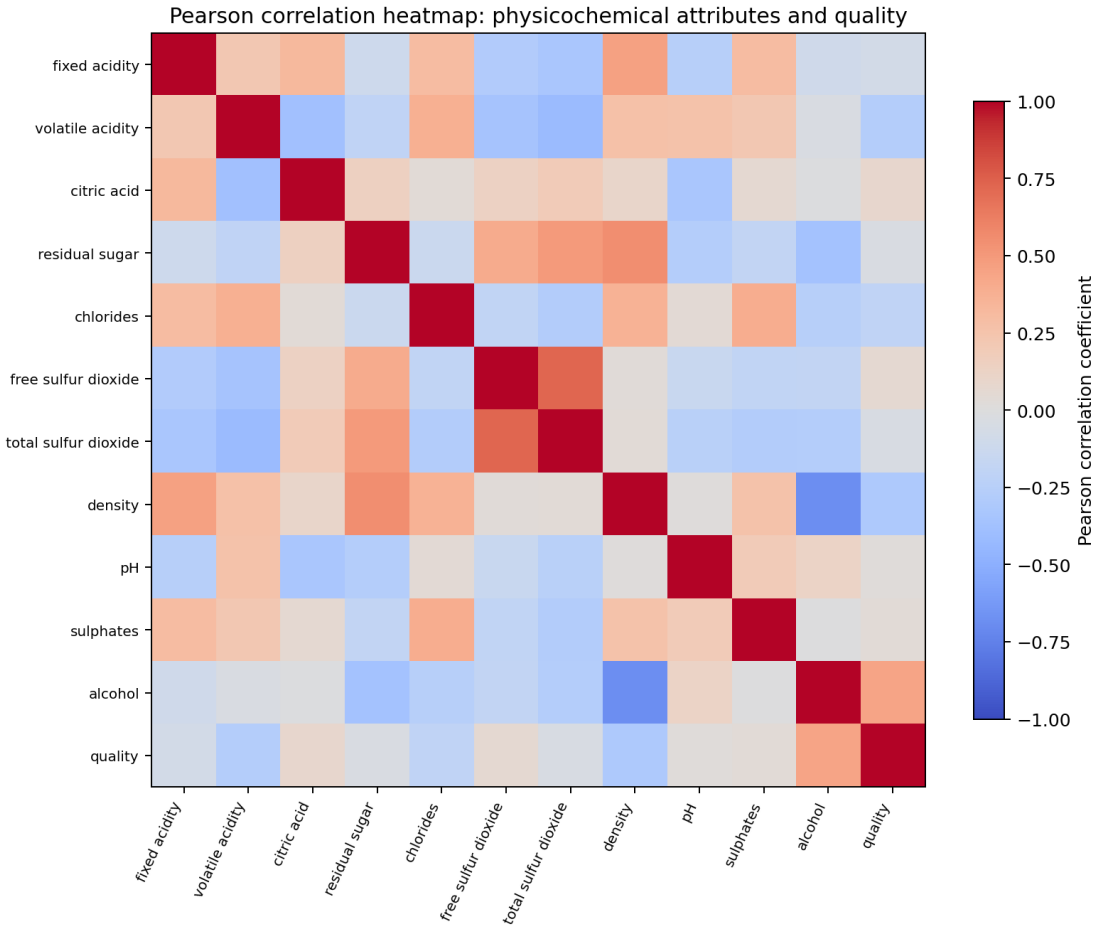
Correlation heatmap and two strongest associations

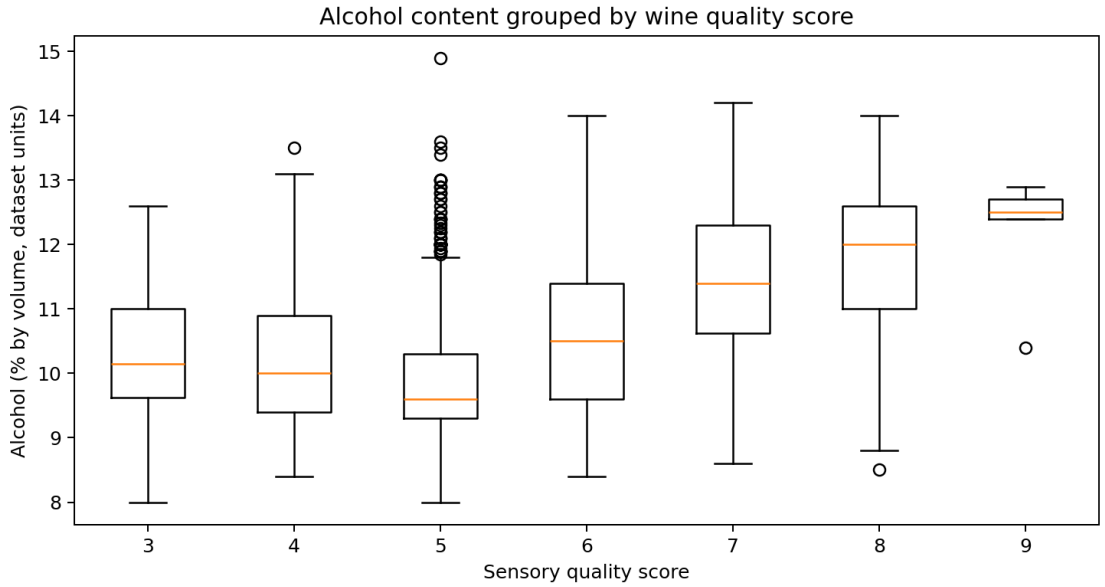
Figure 3. Among the 11 physicochemical predictors, alcohol has the strongest absolute Pearson correlation with quality (r = +0.444).

Density is second (r = -0.306). These are moderate-to-weak associations rather than near-perfect relationships. Density is also

correlated with other predictors, including alcohol (r = -0.684 in the training data) and residual sugar (r = +0.563), so its marginal

correlation should not be interpreted in isolation.





5. Task 3 - Regression model evaluation

A multiple linear regression model was fit with all 11 physicochemical variables. The split produced 5,197 training

records and 1,300 test records; `random_state = 42` was fixed for reproducibility. Predictors were standardized using

the training set before model fitting.

Held-out test metrics

Metric Test value Interpretation

MAE 0.5659 Average absolute prediction error is about 0.57 quality points.

MSE 0.5467 Mean squared error; larger errors receive greater penalty.

RMSE 0.7394 Typical error scale is about 0.74 quality points.

R-squared 0.2598 The model explains about 26.0% of test-set quality variance.

The test R-squared of 0.260 indicates limited predictive power. An MAE of about 0.566 is below one full quality point

on average, but the score range is discrete and high/low grades are rare; average error does not guarantee reliable

decisions on individual samples.

Standardized regression coefficients

Rank Attribute Standardized coefficient

1 alcohol +0.3224

2 residual sugar +0.2204

3 volatile acidity -0.2190

4 density -0.1773

5 total sulfur dioxide -0.1495

6 free sulfur dioxide +0.1225

7 sulphates +0.1202

Alcohol is the largest positive standardized coefficient (+0.322). Residual sugar (+0.220) and volatile acidity (-0.219) have similar

absolute coefficient sizes but opposite signs after adjustment for the other model inputs. These coefficients describe conditional

linear associations, not causal effects.

Multicollinearity diagnostic

Predictor VIF Pair of predictors Pearson r

density 15.51 free sulfur dioxide / total sulfur dioxide +0.724

residual sugar 7.20 density / alcohol -0.684

fixed acidity 4.74 residual sugar / density +0.563

alcohol 4.62 residual sugar / total sulfur dioxide +0.492

total sulfur dioxide 2.96

Density has a VIF of about 15.51, and residual sugar has a VIF of about 7.20, indicating substantial overlap with other

predictors. The largest pairwise correlation shown is between free and total sulfur dioxide (r = +0.724); density and

alcohol are also correlated (r = -0.684). Multicollinearity can make coefficient magnitudes and signs less stable even

when overall predictions remain usable.

6. Prediction quality and business decision-making

Figure 4. Test-set actual versus predicted scores. Actual test scores range from 3 to 9; predictions range from 4.05 to 7.72.

Predictions cluster around the middle of the range and do not reach the rare observed extremes, consistent with the moderate

MAE/RMSE and low R-squared. A linear model tends to regress uncommon scores toward the center.

Business recommendation

Alcohol and density deserve attention during screening because they have the two strongest absolute Pearson

correlations with quality in the pooled data. Density should be interpreted alongside other measurements due to its

multicollinearity with alcohol and residual sugar. Screening thresholds should be validated against sensory scores

before the company uses them for quality decisions.

Can the model replace sensory evaluation?

No. The test-set R-squared of 0.260 shows that most variation in sensory quality remains unexplained, and the

prediction scatter shows limited reproduction of rare high and low scores. The model can prioritize samples for review

or serve as an assistive signal, but replacing trained sensory evaluation is not supported by these results.

Limitations

- The red and white wine datasets are pooled, but wine type is not included as a predictor as requested; the larger

white-wine group dominates pooled statistics.

- Quality scores are imbalanced; only 5 samples have score 9, and extreme scores are poorly represented.
- Linear regression cannot capture nonlinear relationships or interactions unless explicitly engineered.
- Density and other predictors are correlated, which limits isolated interpretation of coefficients.

Conclusion

Alcohol is moderately positively associated with quality, density is moderately negatively associated, and residual

sugar has the greatest relative variability among the three required attributes. The linear regression has moderate

average error but explains only about one quarter of test-set variance. Laboratory measures can support triage and

monitoring; sensory evaluation remains necessary for final quality decisions.

Dataset citation supplied in winequality.names: Cortez, P., Cerdeira, A., Almeida, F., Matos, T., and Reis, J. (2009). Modeling wine

preferences by data mining from physicochemical properties. Decision Support Systems, 47(4), 547-553. DOI: 10.1016/j.dss.2009.05.016.

Execution url: https://colab.research.google.com/drive/1KeZ5ztC0eKBQCzJ6iWKDfdcCg1LNATSH?usp=sharing